

Value-at-Risk Estimation using Extreme Value Analysis

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Introduction

Value-at-Risk (VaR) measures the potential loss in value of a risky asset or portfolio over some defined period with a given distribution.

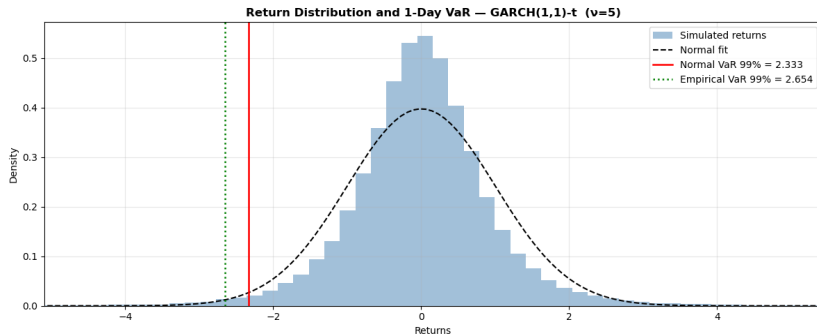
Let S_t be the price of an asset, r_t — its return, and R the log-return over the horizon of interest with cdf F_R . Given the confidence level p , the one-day VaR is the smallest loss level exceeded with probability $1 - p$:

$$\mathbb{P}(r_{t+1} < -\text{VaR}_p | \mathcal{F}_t) = 1 - p, \quad (1)$$

equivalently $\text{VaR}_p = -q_{1-p}(r_{t+1})$, the negative $(1 - p)$ -quantile of the return distribution.

Motivation

VaR is a quantile of the loss distribution. This quantile usually sits in a region where **almost no data is observed**.



- ▶ Returns simulated from GARCH(1,1)- t with $\nu = 5$; the Normal density fitted to the same sample is too thin in the tails.
- ▶ Normal VaR **underestimates** the potential losses.

Modelling returns: GARCH

Volatility of returns is time-dependent [Bollerslev, 1986]. All specifications share the mean equation

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t | \mathcal{F}_{t-1} \sim (0, \sigma_t^2), \quad (2)$$

$$\varepsilon_t = z_t \sigma_t, \quad z_t | \mathcal{F}_{t-1} \sim (0, 1)$$

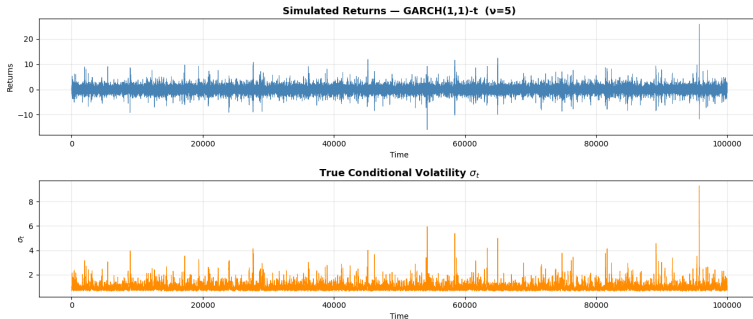
and we consider only the *stationary* case.

► **GARCH(1,1) — Normal:** $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, $z_t | \mathcal{F}_{t-1} \sim \mathcal{N}(0, 1)$; stationarity $\alpha + \beta < 1$.

► **GARCH(1,1) — Student-t:** same variance equation with

$z_t = \frac{Z}{\sqrt{V/\nu}} \cdot \sqrt{\frac{\nu-2}{\nu}}$, where $Z \sim \mathcal{N}(0, 1)$, $V \sim \chi^2(\nu)$; the scaling ensures $\text{Var}(z_t) = 1$ and $\nu > 4$ gives finite kurtosis.

Simulated data



Returns are simulated from GARCH(1,1)- t with $\nu = 5$, and then calibrated to a GARCH(1,1)-Normal model with constant mean by maximum likelihood. The fitted model is deliberately misspecified: whatever tail behaviour remains has to be recovered from the standardised residuals.

VaR estimation: the common scheme

Assume returns follow a GARCH model, $r_t = \mu + \sigma_t z_t$. Then by the definition of VaR,

$$\text{VaR}_p = -\hat{\mu} + \hat{\sigma}_{t+1} q_p, \quad (3)$$

where q_p is the estimated p -quantile of the loss distribution $L = -z$.

The question that motivates EVA

Everything reduces to estimating one number: the **high quantile** q_p of the standardised loss distribution. Fitting Normal quantiles is simple but underestimates VaR because of thin tails.

We compare five estimators of q_p : Normal, Student- t , GEV (Pickands), GPD (joint MLE) and a GPD+GEV hybrid.

GEV approach

Fisher–Tippett theorem

If there exist norming constants $c_n > 0$, $d_n \in \mathbb{R}$ and a non-degenerate df H such that

$$c_n^{-1}(M_n - d_n) \xrightarrow{d} H, \quad (4)$$

then H is of the type of the Fréchet, Weibull or Gumbel df.

Introducing ξ : $\xi = \alpha^{-1} > 0$ for Fréchet, $\xi = 0$ for Gumbel, $\xi = -\alpha^{-1} < 0$ for Weibull, the Jenkinson–von Mises representation collects all three into

$$H_\xi(x) = \begin{cases} \exp\{-(1 + \xi x)^{-1/\xi}\} & \text{if } \xi \neq 0, \\ \exp\{-\exp\{-x\}\} & \text{if } \xi = 0, \end{cases} \quad 1 + \xi x > 0. \quad (5)$$

GEV describes the limit distribution of **normalised maxima**.

Estimating the tail index ξ

Let $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$ be descending order statistics of the losses. The **Pickands estimator** is

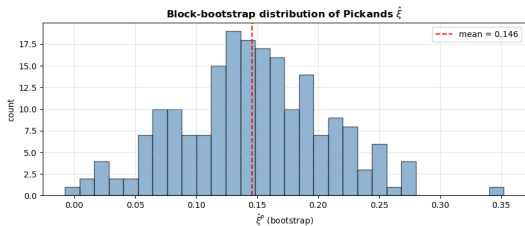
$$\hat{\xi}_k^P = \frac{1}{\log 2} \log \left(\frac{X_{(k)} - X_{(2k)}}{X_{(2k)} - X_{(4k)}} \right). \quad (6)$$

We focus on Pickands rather than Hill: it is consistent for all $\xi \in \mathbb{R}$, whereas Hill is valid only for $\xi > 0$.

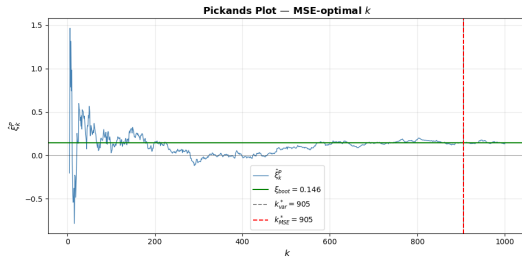
Choosing k :

- ▶ k too large $\Rightarrow$ the estimator moves into the *non-tail* part of the sample and $\hat{\xi}$ is biased. We cap k at **2.5%** of the sample size.
- ▶ The true ξ is unknown, so a proxy $\bar{\xi}_{\text{boot}}$ is obtained by **block bootstrap**.
- ▶ On the full sample, k^* is selected by minimising the MSE of $\hat{\xi}_k^P$ relative to $\bar{\xi}_{\text{boot}}$ over a sliding window W of the Pickands plot.

Choosing k^*



Block-bootstrap distribution of $\hat{\xi}^P$



Pickands plot with MSE-optimal k

GEV quantile

Given the estimated tail index $\hat{\xi}$, the p -quantile can be estimated **outside the range of the observed data**:

$$\hat{x}_{p,k,n} = \frac{\left(\frac{k}{n(1-p)}\right)^{\hat{\xi}} - 1}{1 - 2^{-\hat{\xi}}} (X_{k,n} - X_{2k,n}) + X_{k,n}, \quad (7)$$

and VaR follows from (3).

GPD approach

The generalised Pareto distribution $G_{\xi,\beta}$ with shape $\xi \in \mathbb{R}$ and scale $\beta > 0$ has tail

$$\overline{G}_{\xi,\beta}(x) = \begin{cases} \left(1 + \xi \frac{x}{\beta}\right)^{-1/\xi} & \text{if } \xi \neq 0, \\ e^{-x/\beta} & \text{if } \xi = 0, \end{cases} \quad x \in D(\xi, \beta). \quad (8)$$

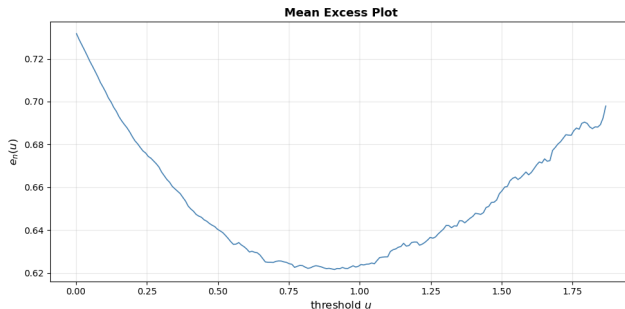
Unlike GEV, the GPD is the limit distribution of **scaled excesses over high thresholds** — peaks over threshold rather than block maxima.

The threshold is chosen from the **mean excess function**

$$e(u) = \mathbb{E}[X - u \mid X > u], \quad u \geq 0, \quad (9)$$

picking u^* so that the empirical $e_n(x)$ is approximately linear for $x \geq u$.

Threshold selection



The mean excess plot flattens and turns linear around $u \approx 1$, so we set $u^* = 1$.

GPD: maximum likelihood estimation

Given the exceedances $Y_1, \dots, Y_N$ over the threshold u , we treat them as an iid sample from a GPD $G_{\xi, \beta}$ and estimate **both** parameters by maximum likelihood:

$$\ell(\xi, \beta) = -N \ln \beta - \left(\frac{1}{\xi} + 1 \right) \sum_{i=1}^N \ln \left(1 + \frac{\xi Y_i}{\beta} \right).$$

Under the reparametrisation $\tau = -\xi/\beta$ [Smith, 1987] the profile MLE of the shape given τ is closed-form, and the remaining score equation is scalar:

$$\hat{\xi}(\tau) = \frac{1}{N} \sum_{i=1}^N \ln(1 - \tau Y_i), \quad h(\tau) = \frac{1}{\tau} + \frac{1}{N} \left(\frac{1}{\hat{\xi}(\tau)} + 1 \right) \sum_{i=1}^N \frac{Y_i}{1 - \tau Y_i} = 0.$$

Solving $h(\tau) = 0$ for $\hat{\tau}$ gives the joint MLE via $\hat{\xi} = \hat{\xi}(\hat{\tau})$, $\hat{\beta} = -\hat{\xi}/\hat{\tau}$: the two-dimensional optimisation is reduced to one-dimensional root-finding.

GPD: Pickands–Balkema–de Haan theorem

For a rv X with df F and right endpoint x_F , and fixed $u < x_F$, the **excess df** over the threshold u is $F_u(x) = P(X - u \leq x \mid X > u)$, $x \geq 0$.

Theorem [Balkema, de Haan, 1974; Pickands, 1975]

For every $\xi \in \mathbb{R}$, $F \in \text{MDA}(H_\xi)$ **if and only if**

$$\lim_{u \uparrow x_F} \sup_{0 < x < x_F - u} |F_u(x) - G_{\xi, \beta(u)}(x)| = 0 \quad \text{for some positive function } \beta.$$

- ▶ Excesses over a high threshold are approximately GPD *exactly* when normalised maxima are approximately GEV — and with **the same** ξ .
- ▶ We therefore take $\hat{\xi}$ from Pickands and maximise the GPD log-likelihood with respect to β only.

GPD quantile

With $(\hat{\xi}, \hat{\beta})$ in hand, the high quantile of the GPD is

$$\hat{x}_p = u + \frac{\beta}{\xi} \left(\left(\frac{n}{N}(1-p) \right)^{-\xi} - 1 \right), \quad \xi \geq 0, \quad (10)$$

where n is the sample size and N the number of exceedances over u , and VaR again follows from (3).

Two routes to the same tail

GEV works with normalised **maxima** and needs a choice of k ; GPD works with **exceedances** and needs a choice of u . The hybrid borrows the shape from the first and the scale from the second.

Backtest

Rolling-window one-day-ahead forecasts, window = 1000, $p = 0.95$ and $p = 0.99$:

1. Fit GARCH(1,1) on the return window, recovering $\hat{\mu}$, $\hat{\sigma}_t$ and the standardised residuals $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$.
2. Forecast next-day volatility $\hat{\sigma}_{t+1}$ (one-step-ahead GARCH forecast).
3. Estimate the p -quantile q_p of the loss distribution $L = -z_t$ — Normal, Student- t , GEV (Pickands), GPD (joint MLE) or the GPD+GEV hybrid.
4. Map back to the return scale: $\text{VaR}_p = -\hat{\mu} + \hat{\sigma}_{t+1} q_p$.

A **violation** occurs when the realised loss exceeds the forecast. The target is the nominal rate itself.

Backtest results

Method	Viol.	Rate	Dev.
Normal	4620	0.04668	0.00332
Student- t	5244	0.05298	0.00298
GEV (Pickands)	5433	0.05489	0.00489
GPD (joint MLE)	5217	0.05271	0.00271
GPD+GEV hybrid	5374	0.05429	0.00429

Table: $p = 0.95$: expected rate 5%

Method	Viol.	Rate	Dev.
Normal	1675	0.01692	0.00692
Student- t	1150	0.01162	0.00162
GEV (Pickands)	1136	0.01148	0.00148
GPD (joint MLE)	1162	0.01174	0.00174
GPD+GEV hybrid	1049	0.01060	0.00060

Table: $p = 0.99$: expected rate 1%

Conclusions and future work

- ▶ EVA estimates the tail from the **extreme observations alone** and extrapolates beyond the largest observed loss.
- ▶ **No single method wins at both levels.** At $p = 0.95$ the joint-MLE GPD is closest to nominal and the hybrid is only fourth; at $p = 0.99$ the order reverses.

Future work:

- ▶ **Continuous-time volatility:** replace the discrete GARCH filter by an SDE-based stochastic volatility model, which supplies $\hat{\sigma}_{t+1}$ while EVA still supplies the tail quantile.
- ▶ **Real market data:** unknown ξ , structural breaks and non-stationarity; Kupiec and Christoffersen tests instead of the raw violation count.
- ▶ **Portfolio VaR.**

Thank you for your attention!